

HR Analytics & Employee Attrition Intelligence Dashboard

Power BI Dashboard Project

Project Overview

The **HR Analytics & Employee Attrition Intelligence Dashboard** is an interactive Business Intelligence solution developed using **Microsoft Power BI** to analyze employee attrition trends and workforce demographics. The dashboard transforms raw HR data into meaningful insights through interactive visualizations, KPI cards, DAX measures, and Power Query transformations.

This project enables HR professionals to identify employee turnover patterns, analyze workforce demographics, and make informed decisions to improve employee retention and organizational performance.

Project Objectives

- Analyze employee attrition trends across the organization.
 - Monitor key HR performance indicators (KPIs).
 - Identify departments and job roles with high employee turnover.
 - Understand the impact of age, gender, and business travel on attrition.
 - Support HR teams in making data-driven decisions.
 - Develop an interactive dashboard using Power BI.
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Business Problem

Employee attrition is a significant challenge for organizations, leading to increased recruitment costs, productivity loss, and operational disruptions.

The primary objective of this dashboard is to:

- Monitor employee attrition rates.
 - Identify departments with high turnover.
 - Analyze workforce demographics.
 - Understand the factors influencing employee attrition.
 - Improve employee retention strategies through data-driven insights.
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Dataset Information

The dataset includes employee-related information such as:

- Employee ID
 - Age
 - Gender
 - Department
 - Job Role
 - Business Travel
 - Monthly Income
 - Attrition Status
 - Education
 - Marital Status
 - Years at Company
 - Job Satisfaction
 - Performance Rating
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Tools & Technologies Used

- Microsoft Power BI
 - Power Query
 - DAX (Data Analysis Expressions)
 - Microsoft Excel
 - Data Modeling
 - Interactive Data Visualization
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Key Performance Indicators (KPIs)

The dashboard provides the following KPIs:

- **Total Employees:** 1,000
- **Average Salary:** 6.50K
- **Average Age:** 36.92 Years
- **Attrition Rate:** 16.08%
- **Attrition Count:** 238

These KPIs provide a quick overview of workforce size, salary distribution, employee demographics, and attrition performance.

Dashboard Features

Interactive Filters (Slicers)

Users can filter the dashboard using:

- Department
- Business Travel
- Age Group
- Gender

The dashboard updates dynamically based on the selected filters, allowing detailed workforce analysis.

Dashboard Visualizations

1. Employee Summary

KPI cards provide an overview of:

- Total Employees
- Average Salary
- Average Age
- Attrition Rate
- Attrition Count

These metrics help HR managers monitor workforce performance at a glance.

2. Department-wise Attrition

A bar chart compares employee attrition across departments.

Departments include:

- Research & Development
- Sales
- Human Resources

Business Insight:

Research & Development experiences the highest employee attrition, while Human Resources records the lowest turnover.

3. Job Role Attrition Analysis

A bar chart displays attrition by job role.

Key job roles include:

- Laboratory Technician
- Sales Executive
- Research Scientist
- Manufacturing Director
- Healthcare Representative
- Manager
- Research Director

Business Insight:

Laboratory Technicians have the highest attrition count, followed by Sales Executives and Research Scientists.

4. Age Group Analysis

Employees are categorized into:

- 18–25
- 26–35
- 36–45
- 46–55
- 55+

Business Insight:

Employees aged **26–35** experience the highest attrition, while employees aged **55+** have the lowest turnover.

5. Gender-wise Attrition

A column or pie chart compares employee attrition by gender.

Business Insight:

Male employees show a higher attrition rate compared to female employees.

6. Business Travel Analysis

The dashboard analyzes employee attrition based on business travel frequency.

Categories include:

- Travel Frequently
- Travel Rarely
- Non-Travel

Business Insight:

Employees who travel frequently tend to exhibit higher attrition compared to non-travel employees.

Data Cleaning & Transformation

The dataset was prepared using **Power Query** by performing the following transformations:

- Removed duplicate records.
 - Handled missing values.
 - Verified and corrected data types.
 - Created age group categories.
 - Standardized column names and data formats.
 - Prepared data for reporting and visualization.
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Data Modeling

The dashboard uses Power BI's data modeling capabilities to establish relationships between tables and support efficient analysis.

The model enables dynamic filtering, fast calculations, and scalable reporting.

DAX Measures

Several DAX measures were created to calculate key metrics, including:

- Total Employees
- Attrition Count
- Attrition Rate
- Average Salary
- Average Age

These measures automatically update based on the selected filters and slicers.

Key Business Insights

The dashboard reveals several valuable insights:

- Research & Development department has the highest employee attrition.
- Sales department experiences the second-highest turnover.

- Human Resources has the lowest attrition.
 - Laboratory Technicians are the most affected job role.
 - Employees aged 26–35 contribute the highest attrition.
 - Employee attrition decreases as age increases.
 - Male employees leave the organization more frequently than female employees.
 - Business travel appears to influence employee turnover.
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Skills Demonstrated

This project demonstrates practical skills in:

- Power BI Dashboard Development
 - Data Cleaning
 - Data Transformation
 - Power Query
 - Data Modeling
 - DAX Calculations
 - HR Analytics
 - Business Intelligence
 - KPI Design
 - Interactive Reporting
 - Data Visualization
 - Analytical Thinking
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Business Questions Answered

This dashboard helps answer the following business questions:

- What is the current employee attrition rate?
 - Which department experiences the highest turnover?
 - Which job roles are most affected by attrition?
 - Which employee age group has the highest attrition?
 - How does attrition vary by gender?
 - Does business travel influence employee turnover?
 - What are the overall workforce demographics?
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Business Impact

The dashboard supports HR professionals by enabling them to:

- Monitor employee turnover trends.
- Identify high-risk employee groups.
- Analyze workforce demographics.

- Improve employee retention strategies.
 - Support strategic HR planning and decision-making.
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Future Enhancements

Future versions of this dashboard may include:

- Employee Attrition Prediction using Machine Learning.
 - Employee Performance Analysis.
 - Department-wise Salary Analysis.
 - Employee Satisfaction Dashboard.
 - Workforce Diversity Metrics.
 - Predictive HR Analytics.
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Conclusion

The **HR Analytics & Employee Attrition Intelligence Dashboard** demonstrates how Microsoft Power BI can transform HR data into actionable business insights. By leveraging Power Query, DAX calculations, interactive visuals, and KPI cards, the dashboard enables HR teams to monitor workforce trends, identify key attrition factors, and make informed decisions that support employee retention and organizational success.

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